

Categories, Proofs, and Processes:

Lecture 8 Equalisers

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1 Feb 2024

3 Limits and Colimits

3.1 Equalisers

3.1.1 Motivation

In **Set**, we can define equality between functions f, g when $\forall x \in X, f(x) = g(x)$, but we might be interested in situations where where $f(x) = g(x)$ for $x \in Y \subsetneq X$.

For example, $X = \{0, 1\}^*$, but only a subset G could encode graphs using this alphabet.

3.1.2 Definition

Definition 3.1. Consider a pair of parallel arrows $A \begin{smallmatrix} \xrightarrow{f} \\ \xrightarrow{g} \end{smallmatrix} B$. An **equaliser** of (f, g) is an arrow $e : E \rightarrow A$ such that

$$f \circ e = g \circ e$$

and for any $h : D \rightarrow A$ such that $f \circ h = g \circ h$, there is a unique $\hat{h} : D \rightarrow E$ so that

$$h = e \circ \hat{h}$$

Diagrammatically,

$$\begin{array}{ccc} E & \xrightarrow{e} & A \\ \uparrow \hat{h} & \nearrow h & \begin{smallmatrix} \xrightarrow{f} \\ \xrightarrow{g} \end{smallmatrix} \\ D & & B \end{array}$$

Note 3.2. A diagram commutes if any two paths with a common source and target, with at least one of the paths at least length > 1 , are equal.

Example 3.3. In **Set**, the equaliser of f, g is given by the inclusion

$$\{x \in A \mid f(x) = g(x)\} \hookrightarrow A$$

For $f, g : \mathbb{R} \rightarrow \mathbb{R}$ where

$$\forall x, f(x) = ax^2 + bx + c, g(x) = 0$$

the equaliser is the roots of $f(x)$ if they exist.

3.1.3 Results

Proposition 3.4. *Any equaliser in any category is monic.*

Proof.

$$\begin{array}{ccccc} E & \xrightarrow{e} & A & \xrightarrow[f]{g} & B \\ x \uparrow \parallel y & \nearrow h & & & \\ D & & & & \end{array}$$

Suppose $e \circ x = e \circ y$, then it is equal to h . Then

$$f \circ h = f \circ e \circ x = g \circ e \circ x = g \circ h$$

There is a unique map $h' : D \rightarrow E$ such that $e \circ h' = h$. But since

$$e \circ x = h \wedge e \circ y = h \implies x = y = h'$$

by uniqueness, so e is monic. □

Definition 3.5. **Coequaliser** in $\mathcal{C}$ is an equaliser in $\mathcal{C}^{op}$.

Proposition 3.6. *Any coequaliser in any category is epic.*

Proposition 3.7. *Equalisers and coequalisers are unique up to a unique isomorphism.*

Proof. Suppose

$$\begin{array}{ccc} E_1 & \xrightarrow{e_1} & A \xrightarrow[f]{g} B \\ & \nearrow e_2 & \\ E_2 & & \end{array}$$

Thus there are unique maps

$$\begin{aligned} u_1 : E_2 &\rightarrow E_1, & e_2 &= e_1 \circ u_1 \\ u_2 : E_1 &\rightarrow E_2, & e_1 &= e_2 \circ u_2 \end{aligned}$$

Then identities exist

$$\begin{aligned} 1_{E_1} : E_1 &\rightarrow E_1 := u_1 \circ u_2 \\ 1_{E_2} : E_2 &\rightarrow E_2 := u_2 \circ u_1 \end{aligned}$$

□

3.2 Pullbacks

3.2.1 Definitions

Definition 3.8. Given a pair of morphisms

$$A \xrightarrow{f} C \xleftarrow{g} B$$

(called a cospan), we define an (f, g) -cone to be

$$A \xleftarrow{p} D \xrightarrow{q} B$$

such that the following diagram commutes.

$$\begin{array}{ccc} D & \xrightarrow{q} & B \\ p \downarrow & & \downarrow g \\ A & \xrightarrow{f} & C \end{array}$$

Definition 3.9. The ***pullback*** of f along g is an (f, g) -cone such that for any other (f, g) -cone $A' \xleftarrow{p'} D' \xrightarrow{q'} B$, there is a unique $h : D' \rightarrow D$ such that $p' = p \circ h, q' = q \circ h$. Diagrammatically,

$$\begin{array}{ccccc} D' & & & & \\ & \searrow^{q'} & & & \\ & & D & \xrightarrow{q} & B \\ & \searrow^{p'} & \downarrow p & & \downarrow g \\ & & A & \xrightarrow{f} & C \end{array}$$

(Note: A dashed arrow h points from D' to D in the original diagram.)

Note 3.10. Pullback often denoted as

$$A \times_C B$$

Example 3.11. In **Set**, pullback is subset of cartesian product. Take a function $f : A \rightarrow B$ and subset $V \subseteq B$. Consider

$$f^{-1}(V) = \{a \in A \mid f(a) \in V\} \subseteq A$$

Consider

$$\begin{array}{ccc} f^{-1}(V) & \xrightarrow{\bar{f}} & V \\ \downarrow & & \downarrow \\ A & \xrightarrow{f} & B \end{array}$$

So $f^{-1}(V)$ is a pullback, unique up to a unique isomorphism.

Proposition 3.12. *Pullbacks are unique up to a unique isomorphism.*

3.2.2 Results

Proposition 3.13. *If a category $\mathcal{C}$ has binary products and equalisers, then it has pullbacks.*

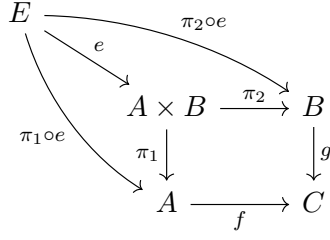
Proof. Given the cospan

$$A \xleftarrow{p} C \xrightarrow{q} B$$

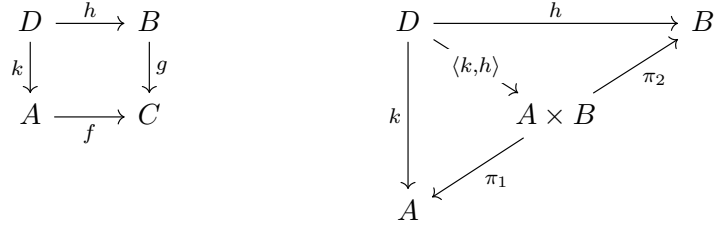
we want to construct its pullback using products and equalisers. Consider the product

$$A \xleftarrow{\pi_1} A \times B \xrightarrow{\pi_2} B$$

Form an equaliser $e : E \rightarrow A \times B$. The following illustrative diagram does not commute



We want to show $A \xleftarrow{\pi_1 \circ e} E \xrightarrow{\pi_2 \circ e} B$ is a pullback, since e might not be unique. Assume the following diagrams commute



Then

$$f \circ \pi_1 \circ \langle k, h \rangle = g \circ \pi_2 \circ \langle k, h \rangle$$

Combine this with the original equaliser and we have

$$\begin{array}{ccccc}
 E & \longrightarrow & A \times B & \xrightarrow[f \circ \pi_1]{g \circ \pi_2} & C \\
 \uparrow u & \nearrow \langle k, h \rangle & & & \\
 D & & & &
 \end{array}$$

Unique map u , thus $A \xleftarrow{\pi_1 \circ e} E \xrightarrow{\pi_2 \circ e} B$ is a pullback. □